

Shingle Scanner - print pack v1

Bulk MTG inventory for the Casa Durval Paiva card drive

What is in this pack

- p.2-4 Column strips, 1:1 scale (2 per page). Columns 1-4 for config C4, 1-6 for C6.
- p.5 Stop cards, 1:1 scale (6 per page). Close any column with fewer than 20 cards.
- p.6 Camera configurations, heights by focal length, and top-view layout.
- p.7 Stand drawing and pin-hole heights (matches shingle_stand.scad).
- p.8 Assembly, phone setup and calibration checklist.

How to print (critical)

1. Print at 100% / 'Actual size'. Turn OFF 'Fit to page', 'Shrink' and borderless scaling.
2. A4 is native. US Letter also works at 100%: all 1:1 content sits inside the Letter area.
3. Measure the 100 mm ruler below and the one on each strip page. Accept 99-101 mm only.
4. Matte paper or cardstock (160-220 g/m²), laser if possible. Glossy paper glares.
5. Glue each strip onto a 70 x 310 mm rigid board (5 mm foam board or 3 mm MDF) to make a tray.

How many copies

C4 (12 MP phones): pages 2-3, three sets = 12 trays | stop cards: page 5, one copy
C6 (24 MP+ phones): pages 2-4, three sets = 18 trays | stop cards: page 5, one copy

The standard (do not change without changing the software)

Card 63 x 88 mm, unsleeved. Offset 10 mm. 20 cards per column. Column pitch = strip width = 70 mm.
Header 25 mm with two 18 mm ArUco markers (DICT_4X4_50). Column j uses ids 2(j-1) and 2(j-1)+1.
Stop card = ArUco id 40. A full column (20 cards) never has a stop card.
Stacking: card 1 at the top line; each next card 10 mm lower and ON TOP. Only card 20 is fully visible.

Side view of a column (not to scale):

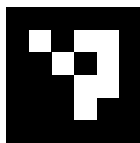
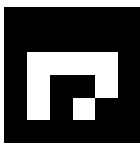
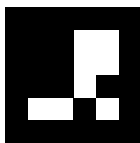
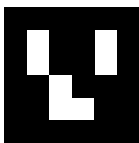


Each card sits 10 mm further along, lying on top of the previous one.

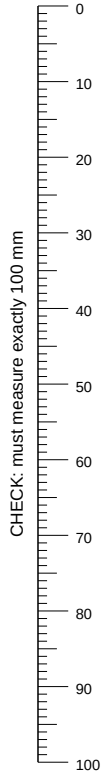


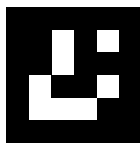
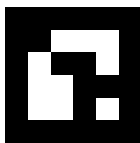
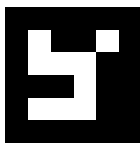
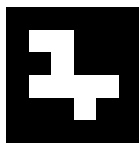
Print check: this ruler must measure exactly 100 mm



COL 1		COL 2	
1	CARD 1 TOP EDGE ON THIS LINE next card 10 mm lower, ON TOP	1	CARD 1 TOP EDGE ON THIS LINE next card 10 mm lower, ON TOP
2		2	
3		3	
4		4	
5		5	
6		6	
7		7	
8		8	
9		9	
10		10	
11		11	
12		12	
13		13	
14		14	
15		15	
16		16	
17		17	
18		18	
19		19	
20		20	
card 20 body continues past strip end Shingle strip v1 col 1 ArUco 4x4_50 ids 0/1		card 20 body continues past strip end Shingle strip v1 col 2 ArUco 4x4_50 ids 2/3	

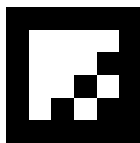
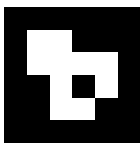
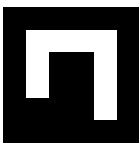
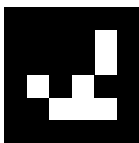
Column strips 1-2 | print at 100% (Actual size) | cut on dashed lines



		COL 3				COL 4	
1	CARD 1 TOP EDGE ON THIS LINE next card 10 mm lower, ON TOP		1	1	CARD 1 TOP EDGE ON THIS LINE next card 10 mm lower, ON TOP		1
2			2	2			2
3			3	3			3
4			4	4			4
5			5	5			5
6			6	6			6
7			7	7			7
8			8	8			8
9			9	9			9
10			10	10			10
11			11	11			11
12			12	12			12
13			13	13			13
14			14	14			14
15			15	15			15
16			16	16			16
17			17	17			17
18			18	18			18
19			19	19			19
20			20	20			20
card 20 body continues past strip end			card 20 body continues past strip end				
Shingle strip v1 col 3 ArUco 4x4_50 ids 4/5			Shingle strip v1 col 4 ArUco 4x4_50 ids 6/7				

Column strips 3-4 | print at 100% (Actual size) | cut on dashed lines

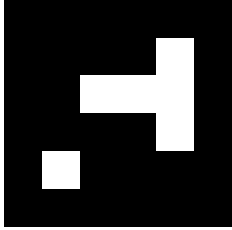


		COL						COL			
		5						6			
1	CARD 1 TOP EDGE ON THIS LINE next card 10 mm lower, ON TOP				1	1	CARD 1 TOP EDGE ON THIS LINE next card 10 mm lower, ON TOP				1
2					2	2					2
3					3	3					3
4					4	4					4
5					5	5					5
6					6	6					6
7					7	7					7
8					8	8					8
9					9	9					9
10					10	10					10
11					11	11					11
12					12	12					12
13					13	13					13
14					14	14					14
15					15	15					15
16					16	16					16
17					17	17					17
18					18	18					18
19					19	19					19
20					20	20					20
-----							-----				
card 20 body continues past strip end							card 20 body continues past strip end				
Shingle strip v1 col 5 ArUco 4x4_50 ids 8/9							Shingle strip v1 col 6 ArUco 4x4_50 ids 10/11				

Column strips 5-6 | print at 100% (Actual size) | cut on dashed lines

Stop cards (1:1) | cut out, glue onto a spare basic land, keep 6+ per scanning station

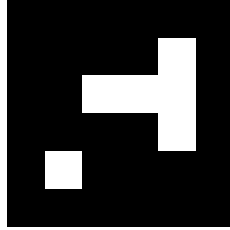
Place a stop card as the next card after the last card of a column that holds fewer than 20.



FIM

END OF COLUMN

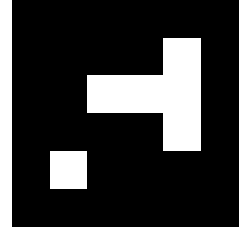
stop card | ArUco 4x4_50 id 40
marker must stay at the TOP



FIM

END OF COLUMN

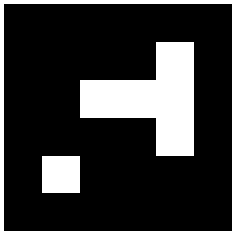
stop card | ArUco 4x4_50 id 40
marker must stay at the TOP



FIM

END OF COLUMN

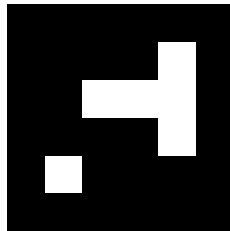
stop card | ArUco 4x4_50 id 40
marker must stay at the TOP



FIM

END OF COLUMN

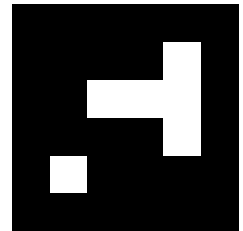
stop card | ArUco 4x4_50 id 40
marker must stay at the TOP



FIM

END OF COLUMN

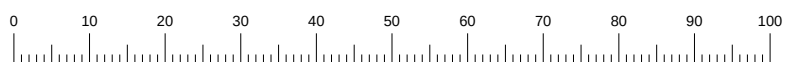
stop card | ArUco 4x4_50 id 40
marker must stay at the TOP



FIM

END OF COLUMN

stop card | ArUco 4x4_50 id 40
marker must stay at the TOP



Print check: this ruler must measure exactly 100 mm

Camera configurations

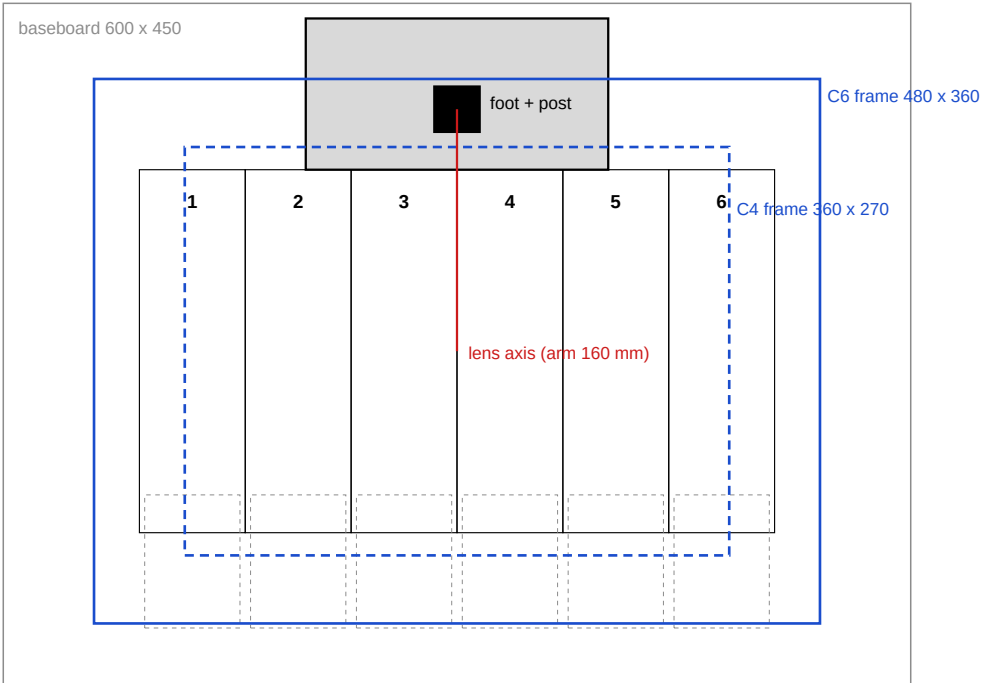
Config	Min. SAVED resolution	Frame on mat (4:3)	px/mm at min.	Cards per photo
C4	12 MP (4032 x 3024)	360 x 270 mm	11.2	4 columns x 20 = 80
C6	24 MP (5712 x 4284)	480 x 360 mm	11.9	6 columns x 20 = 120

Long side of the frame runs ACROSS the columns. Target ≥ 11 px/mm, hard floor 10 px/mm (name text ~ 24 px).
A 12 MP phone cannot run C6 (8.4 px/mm). A 48/50 MP phone saving 12 MP by default counts as 12 MP.
Height (mat surface to phone back) = $W \times f / 34.6$ mm, W = frame long side, f = main lens 35 mm-equivalent.

f (35mm eq.)	C4 lens height	C6 lens height
23 mm	240 mm (exact 239)	320 mm (exact 319)
24 mm	250 mm (exact 250)	330 mm (exact 333)
25 mm	260 mm (exact 260)	350 mm (exact 347)
26 mm	270 mm (exact 270)	360 mm (exact 361)
27 mm	280 mm (exact 281)	370 mm (exact 374)
28 mm	290 mm (exact 291)	390 mm (exact 388)

Start from EXIF 'FocalLengthIn35mmFilm'; the calibration photo is the final authority (p.8).
Phone: main 1x lens only, macro auto-switch OFF, 4:3, highest native resolution, focus + exposure locked, flash off, remote / 2 s timer shutter. Light: two diffuse lamps at ~ 45 deg on opposite sides, no overhead glare.

Top view, scale 1:5 (NOT for measuring)

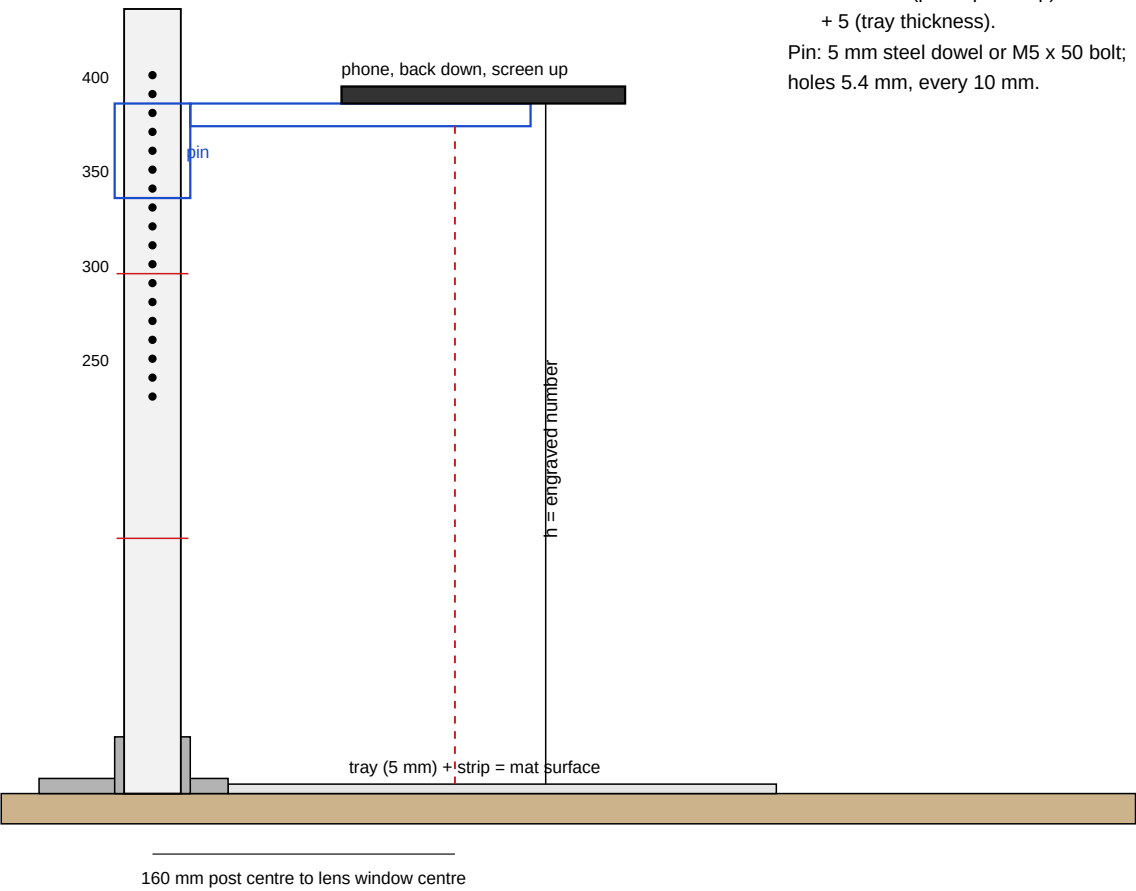


C6: 6 trays, gap $\frac{3}{4}$ on the foot's centre notch. C4: 4 trays, gap $\frac{2}{3}$ on the notch.
Card 20 bodies (dashed) hang past the strip ends onto the baseboard.

Stand: drawing and pin heights

h (lens, mm)	pin z above board	used by
230	210 mm	
240	220 mm	C4-23
250	230 mm	C4-24
260	240 mm	C4-25
270	250 mm	C4-26
280	260 mm	C4-27
290	270 mm	C4-28
300	280 mm	
310	290 mm	
320	300 mm	C6-23
330	310 mm	C6-24
340	320 mm	
350	330 mm	C6-25
360	340 mm	C6-26
370	350 mm	C6-27
380	360 mm	
390	370 mm	C6-28
400	380 mm	

Engraved number = lens height above the MAT, assuming 5 mm trays.
Pin z = h - 25 (pin to plate top) + 5 (tray thickness).
Pin: 5 mm steel dowel or M5 x 50 bolt; holes 5.4 mm, every 10 mm.



Side view, scale 1:4 (NOT for measuring). Red lines = post segment joints (135, 275 mm). Carriage shown at h = 360 mm (C6, 26 mm lens).

Assembly, phone setup and calibration

Parts (see `shingle_stand.scad`, parameter 'part')

foot x1 (200 x 100 x 8 mm + socket boss) | post_1, post_2, post_3 (30 x 30 mm, pegged joints)

carriage x1 (sleeve 50 mm, arm 160 mm, 120 x 120 mm plate, 70 x 70 mm lens window)

5 mm pin | 4 wood screws for the foot | 600 x 450 mm baseboard | rubber bands for the phone

Print: PLA/PETG, 0.2 mm layers, 4 walls, 25% infill. Posts upright; carriage plate-side down.

Assembly

1. Screw the foot to the baseboard, front edge 110 mm from the board's back edge, centred.
2. Stack post_1..post_3 in the socket (pegs up). A pin through a joint hole also locks that joint.
3. Slide the carriage on, arm pointing over the board. Pin it at the hole from the table on p.6/p.7.
4. Draw the tray line: trays butt against the foot's front edge, middle gap on the centre notch.

Phone setup (every session)

- ☐ main 1x lens, macro auto-switch OFF ☐ 4:3, highest native resolution
- ☐ focus + exposure locked on the mat ☐ flash OFF, HDR/night mode OFF if possible
- ☐ remote / earbuds / 2 s timer shutter ☐ main lens centred on the plate crosshair

Calibration (once per phone; repeat if the stand moves)

1. Place all trays for the config, every slot filled with cards.
2. Take one photo. Run: `shingle calibrate photo.jpg --config C4` (or C6)
3. The tool checks: every marker id found and ≥ 5 mm inside the frame; $\text{px/mm} \geq 11$;
tilt: marker scale varies $< 3\%$ across the frame; sharpness per column above threshold.
4. Result: OK / raise one hole / lower one hole / re-centre phone. Save the OK result as the phone profile.

Per-photo cycle (camera operator)

1. Box set in the app. 2. Trays in, against the foot. 3. Shoot. 4. Wait for green 'all columns found'.
5. Square up each column toward its header, face up, in order; file behind the next numbered divider.

Column rules

Exactly 20 cards, OR fewer followed by a stop card (only at the end of a box).

Card top edges on the tick lines; ± 2 mm is fine, but never cover a name bar.

Unsleeved only. Flatten badly curled cards or put them in the manual pile.